Validation report

Product: Institute Intelligence — Case B “The Data Moat”
Version: pilot-1.0
Date: 23 October 2026
Status: prototype validated on synthetic data; ready for a controlled design-partner pilot

Executive summary

The prototype works end to end: 3 synthetic institutes, 1,000 students, 181,201 responses and 52,000 attendance records flow through CSV ingestion, Rasch mastery, three risk targets, intervention tracking, bilingual reports and parent links.

- **Functional:** 6/6 automated tests pass; all primary pages return 200; import, commit, report and parent-link flows completed.
- **Model:** Rasch converges in 41–76 iterations. Disengagement AUC is 0.92–1.00 on held-out synthetic splits. Internal percentile coverage is 74–90% against a nominal 80%; MAE is 8.1–8.5 marks.
- **Privacy:** tenant isolation, consent gating, teacher fee redaction and parent no-risk/no-fee views were verified.
- **Performance:** dashboard p50 99 ms and p95 257 ms under 100 requests at 10-way concurrency; full analytics rebuild for 181k responses takes 7.7 s.
- **Caveat:** this is synthetic data with a designed dropout signal. The AUC is not a real-world claim. The next step is a time-based, two-institute pilot.

1. Scope of validation

In scope	Out of scope
CSV import, validation, quarantine, commit	Native Google Forms connector
Rasch and topic mastery	Production OMR scanner
Disengagement, falling-behind, fee-default scores	Real payment collection
Internal percentile band	JEE/NEET rank prediction
English/Hindi report, parent link	Live WhatsApp provider integration
RLS, roles, consent, audit, erasure request	Legal sign-off under DPDP
Synthetic pilot at 3 institutes	Real institute outcomes

2. Environment and data

Item	Value
Application	FastAPI + Jinja, Python 3.14
Database	PostgreSQL 16.15, Docker, port 55432
Tenants	3: Patna NEET Academy (400), Gaya Success Point (350), Patna JEE Circle (250)
Students	1,000
Tests	55 (54 seeded + 1 imported)
Items	510
Responses	181,201
Attendance records	52,000
Reports	30 per institute, English + Hindi
As-of date	30 September 2026
Raw import	1,205 rows: 1,200 accepted, 4 quarantined, 1 duplicate

The synthetic generator creates ability, topic affinity, attendance decline and withdrawal. That makes the data useful for testing plumbing and failure modes, but it also makes the risk metrics optimistic.

3. Functional validation

ID	Test	Method	Result
V1	Tenant isolation	pytest: count students per tenant; bare connection	Pass: 350 / 250 / 400; bare connection 0
V2	Cross-tenant visibility	tenant A query filtered by tenant B	Pass: 0 foreign rows
V3	Auth and redirects	pytest + TestClient	Pass: unauth 303; bad login 200; good login 303
V4	Teacher fee redaction	Playwright text scan of dashboard/student	Pass: no ₹ amounts; fee-default target hidden
V5	Parent privacy	Playwright parent link	Pass: report visible; no risk label or fee amount
V6	Consent enforcement	revoke `parent_communication`, generate report	Pass: `{"error": "consent"}`
V7	CSV validation	1,205-row demo file	Pass: 1,200 accepted / 4 quarantined / 1 duplicate
V8	Idempotent upload	same content hash twice	Pass: same job id, `existing=True`
V9	Commit	import job commit	Pass: 1,200 responses, 1 test, 0 new students
V10	WhatsApp entry	signed webhook, roll/test/question/answer	Pass: response stored as `form`, confidence 0.80, marks 4
V11	WhatsApp report queue	report page action	Pass: `message_log` row queued with utility template
V12	Erasure request	owner action on student	Pass: pending request + audit event
V13	Page health	owner session, 6 pages	Pass: 200 for dashboard, students, mastery, risk, import, report, status
V14	Automated suite	`python3 -m pytest -q`	Pass: 6/6

Import evidence

```
total: 1,205
accepted: 1,200
quarantined: 4
duplicate: 1
commit: inserted 1,200; duplicate 0; tests_created 1; students_created 0
```

The four quarantine rows exercise unknown topic code, malformed date, missing answer key and out-of-range question number. These are visible first in the review screen.

4. Model validation

4.1 Rasch / mastery

Institute	Students	Items	Responses	θ sd	b sd	Iterations
Patna NEET Academy	400	169	69,599	0.718	0.829	46
Gaya Success Point	350	140	59,818	0.732	0.899	76
Patna JEE Circle	250	146	42,781	0.716	0.887	41

θ sd near 0.7 is plausible for a mixed Class 11/12/dropper cohort. The item spread (b sd 0.83–0.90) means the tests contain a usable range of difficulty. Convergence is stable; no divergence was observed after damping and recentering.

4.2 Disengagement (28-day)

Institute	Positives	Base rate	AUC	PR-AUC	Brier	Recall @10%	Precision @10%
Patna NEET Academy	24	6.0%	0.992	0.925	0.035	1.00	0.63
Gaya Success Point	19	5.4%	1.000	1.000	0.016	1.00	0.57
Patna JEE Circle	15	6.0%	0.922	0.738	0.025	0.67	0.40

Interpretation. The model ranks synthetic withdrawals well, especially in Patna NEET. Gaya's perfect AUC reflects a clean synthetic separation, not model superiority. JEE Circle is the most honest stress case: 15 positives, AUC 0.922, top-10% recall 0.67. The pilot must replace this with time-based validation on real withdrawals.

4.3 Internal percentile band

Institute	n	MAE (marks)	80% interval coverage	q80 absolute error
Patna NEET Academy	400	8.07	90.0%	15.6
Gaya Success Point	350	8.38	74.3%	12.3
Patna JEE Circle	250	8.47	84.0%	14.6

Nominal coverage is 80%. The observed range (74–90%) is acceptable for a pilot but shows that the interval is not perfectly calibrated across cohorts. Reports label the interval as a range and explicitly say it is not a JEE/NEET rank.

4.4 Rule-based targets

Institute	Falling behind high/medium	Fee default high/medium
Patna NEET Academy	48 / 122	20 / 136
Gaya Success Point	9 / 88	12 / 123
Patna JEE Circle	19 / 64	4 / 92

These are deliberately rules, not models. They are explainable and tunable, but they are not validated predictors. Fee amounts are visible only to owner/academic head/admin.

5. Privacy and security validation

Control	Evidence	Verdict
RLS	per-tenant counts; no-context returns 0; cross-tenant 0	Pass
Session	HMAC-signed cookie; 12-hour expiry	Pass
Passwords	scrypt, constant-time verification	Pass
CSRF	state-changing forms include session token	Pass
CSP/headers	frame-deny, nosniff, same-origin	Pass
Parent token	SHA-256 stored; raw token only in link; 90-day expiry	Pass
Consent gate	report and parent link blocked after withdrawal	Pass
Teacher redaction	no ₹ on teacher dashboard/student; fee-default hidden	Pass
Parent content	no risk labels or fee amounts	Pass
Audit	report, link, import, intervention, erasure events	Pass
Erasure	pending request with 30-day workflow	Pass (workflow stub)

Residual risks. The WhatsApp provider is stubbed. The parent link is bearer-token based; anyone with the link can view it until expiry. Production needs rate limiting, link revocation UI and a shorter default lifetime. The dev database uses a known password and should never be deployed as-is.

6. Performance and reliability

Metric	Result	Method
Dashboard latency	74 ms	single authenticated request
Students page	58 ms	200 rows
Mastery grid	88 ms	300 KB HTML
Risk page	34 ms	top 120 rows
Status page	27 ms	model cards
Dashboard p50 / p95	99 / 257 ms	100 requests, 10 workers
Analytics rebuild	7.7 s	3 institutes, 181k responses
Database size	156 MB	`pg_database_size`

The current build is comfortable for a 3-institute pilot. The mastery page is the heaviest; at 10× students it needs pagination or a materialised grid.

7. Failure modes and mitigations

Failure	Current behaviour	Next step
Bad CSV	row-level quarantine; commit blocked until reviewed	Add downloadable error CSV
Duplicate upload	same job returned by content hash	Show prior commit status clearly
Analytics crash	job transaction rolls back; prior snapshots remain	Add job retry/alerting
Wrong tenant	RLS returns zero rows	Keep tests in CI
Consent withdrawn	report/link generation refused	Add consent expiry reminders
Parent link leaked	bearer token valid until expiry	Add revoke button + shorter TTL
Model drift	versioned snapshots	Monthly backtest + drift alert
Fee data leaked to teacher	role filter + template redaction	Add API-level test
LLM hallucination	template path; fact validator	Keep LLM off until eval set exists

8. Limitations

1. **Synthetic outcomes.** Withdrawal is generated from attendance and fee behaviour, so the model partly learns the generator. AUC 1.0 is not evidence of real-world performance.
2. **Random split, not time split.** The pilot must validate on future cohorts to avoid leakage.
3. **Small positive count.** 15–24 positives per institute gives wide confidence intervals.
4. **Rules are not predictors.** Fee-default and falling-behind scores are triage heuristics.
5. **No real institute data.** No DPDP notice, DPA or parent consent text has been legally reviewed.
6. **No native Forms/OMR.** CSV export is the working path; OMR is an adapter point.
7. **WhatsApp is partial.** Inbound structured entry works; outbound delivery is queued, not sent.
8. **Single region.** Patna/Bihar pricing and language assumptions are untested.

9. Pilot plan

Design: 2 institutes, 8 weeks, 300–600 students, one control batch per institute where feasible.

Hypothesis	Measure	Success bar
H1: flags are actionable	teacher-rated usefulness of top-20 flags	≥70% “actionable”
H2: onboarding is light	upload → first insight	<30 minutes
H3: parents engage	report open / reply rate	>60% open, >15% reply
H4: retention improves	28-day attendance vs control	+3 percentage points
H5: willingness to pay	signed LOI / paid pilot	≥₹35/student/month

Guardrails: no student-facing risk labels; no exam-rank claims; no fee data to teachers; consent before parent communication; weekly model review; kill switch for reports.

Decision gate: proceed to paid rollout only if H1, H2 and H5 pass and no privacy incident occurs.

10. Recommendation

Proceed to a controlled design-partner pilot. The prototype is strong enough to test workflow, trust and willingness to pay. It is not yet evidence that the risk model predicts real dropout. The next validation must be temporal, on real consented data, with a pre-registered success bar.

Appendix A — Reproduce

```
python3 scripts/apply_migrations.py
python3 -m app.synth
python3 -m app.analytics
python3 -m pytest -q
uvicorn app.main:app --port 8000
```

Appendix B — Evidence index

- `tests/test_model.py`: Rasch recovery, mastery bounds.
- `tests/test_integration.py`: RLS, import quarantine, consent, API roles.
- `db/tests/rls_smoke.sql`: original RLS/job-queue smoke test.
- `docs/assets/`: dashboard, student, report, risk, import and status screenshots.
- `docs/institute_intelligence_financial_model.xlsx`: unit economics and 3-year plan.